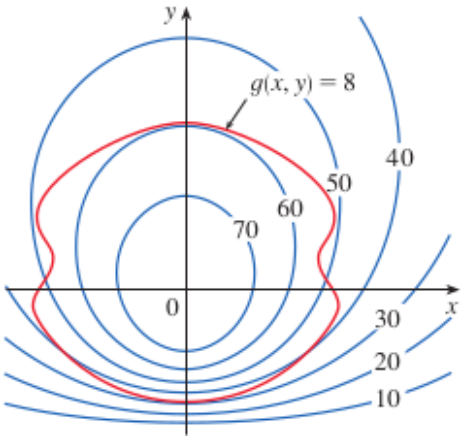
Math 1200 Lecture 10

Oct 6 2026

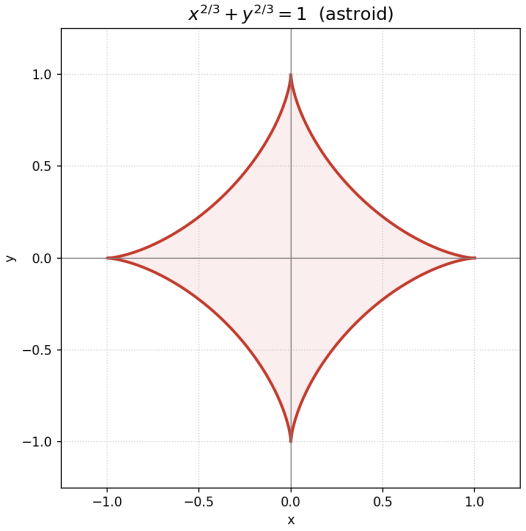
Topics: 14.7: Optimization, Lagrange multipliers Name:_____
Collaborators:_____

Group work 1 The below figure shows a contour plot of a function $f(x, y)$ in blue and the curve $g(x, y) = 8$ in red. Estimate the minimum and maximum value of $f(x, y)$ subject to the constraint $g(x, y) = 8$ using the plot.



Group work 2 Find the minimum and maximum values of the function $f(x, y) = 2x - 2y$ on the solid circle $x^2 + y^2 \leq 4$.

Group work 3 The region $D = \{(x, y) \in \mathbb{R}^2 \mid x^{2/3} + y^{2/3} \leq 1\}$ is plotted below. Find the extreme values of the function $f(x, y) = xy$ on D.



Group work 4 Show using Lagrange multipliers that the rectangle with largest area for a given perimeter is a square.

Group work 5 (Challenge problem) Find the point on the surface $xy^2z^3 = 2$ closest to the origin.